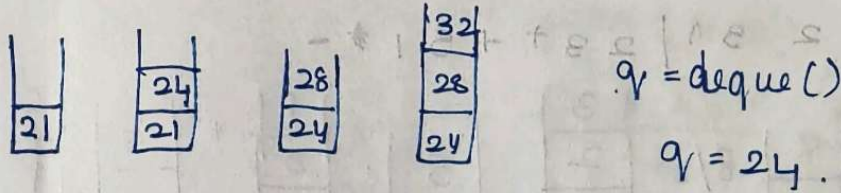
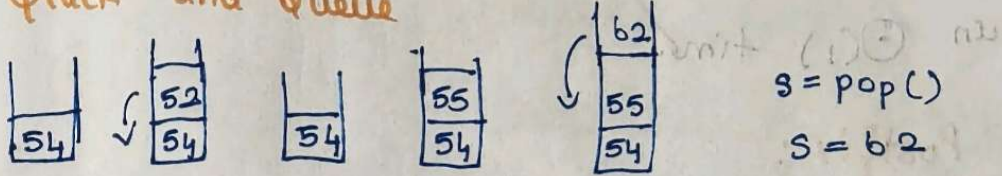


29/7/20

Activity 2

S. Thanishka
192521457

① Stack and Queue



$$s + q = 62 + 24 = 86$$

② Postfix Expression:

$$10\ 5\ +\ 60\ 6\ /\ * 8\ -$$

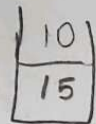
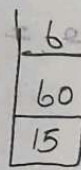
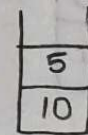
$$10\ 5\ + = 15$$

$$60\ 6\ / = 10$$

$$15 \times 10 = 150$$

$$\text{Answer} : 142$$

10	-	10
5	-	5
+	$10 + 5$	15
60	-	60
6	-	6
/	$6 / 60$	10
*	$10 * 15$	150
-	$150 - 8$	142

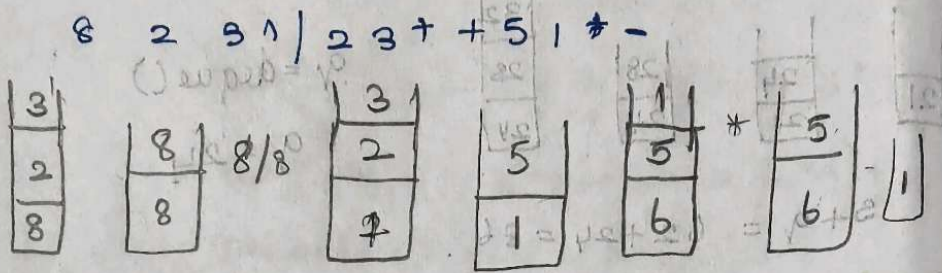


③ The minimum number of Queue operations on Q_1 required to place the elements of Q_1 in Q_2 in reverse order without using any additional storage is 4 enqueue operations

④ Stack with Reverse:

Reverse operation can be implemented in $O(1)$ time.

⑤ Postfix:

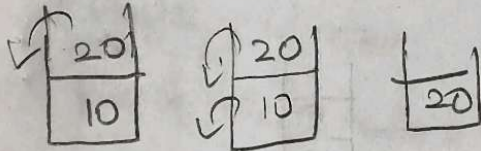


⑥ Infix to Postfix:

$a + b \times c - d \wedge e \wedge f$

Postfix: $a b c * + d e f \wedge \wedge -$

⑦ Stack



Popped sequence = 20, 20, 10, 20

The minimum number of push operations is 4. The elements of the stack are pushed in the order: 20, 20, 10, 20. The popped sequence is 20, 20, 10, 20.